



Outlier detection based on weighted neighbourhood information network for mixed-valued datasets

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ABSTRACT

Outlier detection is of great importance in industry as unexpected errors or faults, abnormal behaviours or phenomena, etc. can occur due to a variety of human, system, and environmental reasons. To identify and analyse these rare items, events or observations can find either anomalies or novelties and, as a result, can help avoid potential unexpected consequences or improve industrial system performance. The operating data collected from industrial systems in the Industry 4.0 era are characterized as multi-attribute (e.g., both numerical and categorical) compared to previous studies. Therefore, a new outlier detection method for mixed-valued datasets based on the weighted network model is proposed in this paper. Concretely, a weighted neighbourhood information network (WNIN) is constructed by considering the neighbourhood relations and similarities among objects to represent a dataset with mixed-valued attributes (DMA). A tailored Markov random walk method is employed to detect outlier on the predefined network model. After reaching the equilibrium, the inlier score is defined according to the out-degree of nodes in the WNIN to represent the inlier degree of objects. Experiments on two real datasets and a case study illustrate the effectiveness and adaptability of the proposed method.

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1. Introduction

The realisation of knowledge discovery in databases (KDD) is an important issue in the development of knowledge-based and data-based systems [42]. Knowledge discovery includes four aspects, i.e., dependency detection, class identification, class description, and outlier detection [16], of which outlier detection aims at identifying and analysing rare observations that are far from the majority of the data [13]. Outliers can be anomalies, novelties, noise, deviations, and exceptions. An outlier usually represents a new perspective or a specific mechanism, and it attracts higher interest than the normal instances. The research on various techniques to implement outlier detection has been widely studied and applied by academia and industry. Therefore, it has been widely used in different domains, e.g., intrusion and fraud detection, medical diagnosis, weather prediction, and social networks [29,18,27,40]; and can also be applied in binary classification, prediction, and event classification [28,23,31].

The approaches for outlier detection can be classified into five categories: distribution-based, depth-based, clustering, distance-based, and density-based approaches. However, these traditional methods usually require additional information or knowledge and can only be applied under specific conditions [42]. In recent years, outlier detection approaches based

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on graphs or networks have attracted increasing interest due to their robust expressiveness and extraordinary ability to capture the long-range correlations within datasets [1,30]. Generally, each data object is modelled as a node in a graph or network, and the relationship between objects is defined as the edge that connects the nodes. The nodes and edges in the graph or network are analysed according to the substructure or correlation to define the outlier score of each object [36]. Zhu et al. [44] defined the natural neighbourhood based on the k -nearest neighbours model and proposed the weighted natural neighbourhood graph model to implement outlier detection with no neighbourhood parameters. Wang et al. [35] constructed weighted directed neighbourhood graphs by using a sequence of automatically determined neighbourhood sizes. A Markov random walk process is performed on the predefined graphs utilizing the different aspects of local information to calculate the outlier score of each object. Most of these approaches can be used to process only numerical data. However, for many real-world problems, the datasets tend to include both numerical and categorical attributes simultaneously; therefore, these datasets are datasets with mixed-valued attributes (DMAs) [17].

The main technological difficulties for outlier detection incorporating a DMA are the measurement of the neighbourhood relations among the objects and the outlier degree of each object. The existing methodologies for measuring neighbourhood relations can be classified into two categories. One category is the k -nearest neighbours in which the nearest k points are determined by calculating and sorting the similarity between the new sample points and all the training samples [9]. In this category, a larger k is used to avoid identifying large numbers of outliers whereas a smaller k is applied to ensure that the points are sufficiently close to the specific object [7]. The other category is the neighbourhood rough set in which the key is to define the neighbours and identify the number (affected by the neighbour radius) of neighbours for each object [32]. In this category, a larger r is used to obtain a better identification result whereas a smaller r is applied to acquire the nearest neighbours. In these methods, the use of a suitable method to measure the similarity is critical to identify whether an object is a neighbourhood of another object. The methods to measure the similarity between the objects in a DMA can be classified into two categories. The main idea of the first category is to convert one type of attribute to another type via transformation (for categorical attributes) or discretisation (for numerical attributes). However, this transformation can result in information loss, changes in the meaning of the data, and an increase in the noise level [26]. The key of the second category is to define a generalised criterion in which the numerical and categorical similarities are defined using the distance function and distribution centroid, respectively [15]. Several useful and valuable trials seeking to measure the outlier degree of an object have been reported in the literature. Moonesignhe and Tan [24] utilised the stationary distribution vector to express the outlier score based on the random walk. Bouguessa [3] proposed the use of the square of the distance between the objects and their k -nearest neighbours in the numerical space and the frequency of the objects in the categorical space to express the outlier score.

Motivated and inspired by the above observations, we introduce a new outlier detection approach, which does not require any prior knowledge, to handle mixed-valued attribute data. A tailored Markov random walk process on a predefined weighted neighbourhood information network (WNIN) is employed to determine the outlier degree for an object. First, the WNIN is modelled based on the neighbourhood information system. Second, the weights of the edges and the state transition probability matrix are obtained according to the similarities and neighbourhood relations among the objects. Then, the outlier score of each object is obtained after the Markov random walk process reaches an equilibrium state. Finally, based on the UCI datasets and a real-world application, the proposed method is compared with other existing outlier detection methods, and the results show that the proposed method is more effective and accurate.

The remainder of this paper is organized as follows. The related work is reviewed in Section 2. The neighbourhood information system and Markov chain are introduced in Section 3. The proposed approach is described in Section 4. The UCI dataset experiments are analysed in Section 5. A real case study is illustrated in Section 6. The conclusions and future work are summarized in Section 7.

2. Literature review

This section provides a review of the studies on outlier detection based on neighbourhood rough sets and network representations for datasets.

The identification methods based on the rough set are completely data-based and never require prior knowledge. In recent years, neighbourhood rough set-based methods have been effectively and widely applied for attribute reduction, feature selection, classification, and gene selection [4,43,6,32]. For outlier detection, some neighbourhood information system-based identification works are mainly restricted to numerical data [5]. Li et al. [19] proposed a novel approach based on the neighbourhood rough set model that can handle real-valued data (uncertainty information) while simultaneously retaining the semantics dataset. As a dataset usually tends to include numerical and categorical attributes, several methods based on the neighbourhood rough set model have been proposed to handle numerical, categorical and hybrid data simultaneously [38]. Goh et al. [11] proposed an anomaly identification approach by using correctness matching based on the neighbourhood rough set, and the method can work well in the case of DMAs and categorical data. In the method proposed in [12], the neighbourhood rough set is modified and applied as a data preprocessing method to select samples, and the samples not selected for training are considered outliers. Zhong et al. [42] researched the hybrid data-driven outlier detection process based on the neighbourhood rough set. This method could handle numerical data, categorical data and mixed-valued data, and the detection results reflected the effectiveness and adaptability of the approach.

Most studies on outlier detection based on the network representation are based on the definition of outliers. Li et al. [22] extracted the features of video images based on Markov random fields to recognise group outlier behaviours. Wang and Davidson [37] addressed the problem involving insufficient a priori contextual information by proposing a probabilistic approach based on random walks, which can simultaneously explore the meaningful contexts and associated score contextual outliers. Real world data are usually multidimensional; thus, Li et al. [21] formulated an algorithm based on the weighted hypergraph model by using the attribute similarity to detect the anomalies in a high-dimensional space. Wang et al. [33] proposed a trajectory-based outlier detection algorithm that involved model training and model-based likelihood estimation based on multidimensional hidden Markov models. In terms of data types, Moonesignhe and Tan [24] presented a stochastic graph-based algorithm for detecting outlying objects with numerical attributes. Wang et al. [34] presented a new outlier detection model that combines the graph representation with the local neighbourhood information around each object to construct a local information graph. This model calculates the outlier score by performing a random walk process on the graph. Soon afterwards, these researchers proposed a new outlier detection model by using a virtual neighbourhood graph [36]. However, both methods can only handle numerical data. In summary, the majority of network-based outlier detection methods focus more on the definition of the outliers and dimensions of the data while datasets with different types of attributes, especially DMAs, are seldom considered.

Motivated by previous researches and emerging new features of datasets, an outlier detection method based on the neighbourhood rough set and network model is proposed in this paper. Three main contributions of this paper can be summarised as follows:

- (1) A WNIN-based outlier detection approach for DMAs is proposed in this study. An improved unified similarity measurement can quantify the comprehensive relations among the objects with mixed-valued attributes. The network representation for a DMA inherits its advantage of interpretability.
- (2) A tailored Markov random walk process in the predefined WNIN is employed to determine the inlier degree for each object. The proposed neighbourhood determination method for an object can decrease the limit of the number of neighbours compared to the k NN-based approach [35,36] and construct a dense network with large connectivity that can ensure the convergence of the random walk process.
- (3) Two experimental and applicable case studies are implemented to verify the utility and effectiveness of the proposed approach. Specifically, a parameter-adjustment logic to reduce the sensitivity of the only adjustable parameter is detailed. Moreover, the identification rationality of the applicable case study is verified from the viewpoint of engineering reality.

3. Preliminaries

3.1. Neighbourhood information system

Let $NIS = (X, C, V, f)$ be a neighbourhood information system, where $X = \{x_1, x_2, \dots, x_n\}$ is a nonempty finite set of objects known as a universe, $C = \{c^1, c^2, \dots, c^m\}$ is a nonempty finite set of attributes, $V = \bigcup_{c \in C} V_c$ is the union of attribute domains, and $f : X \times C \rightarrow V$ is an information function [42]. For a DMA, x_i can be characterized by k numerical attributes and l categorical attributes, where $k + l = m$. The value of the i^{th} object on the h^{th} attribute can be denoted as $x_i^{h'}$ ($i = 1, 2, \dots, n; h = 1, 2, \dots, k, k + 1, \dots, k + l - 1, m$).

The neighbour of x_i on C is defined as:

$$N_i = \{x_j | x_i, x_j \in X, D_{ij} \leq r\} \quad (i, j = 1, 2, \dots, n), \quad (1)$$

where $r \in R^+$ is a neighbourhood radius and D_{ij} is a distance function to calculate the distance between x_i and x_j .

Furthermore, the neighbourhood relation in X is defined as:

$$R = \{(x_i, x_j) \in X \times X | x_j \in N_i\} \quad (i, j = 1, 2, \dots, n). \quad (2)$$

Then, the quotient set X/R constitutes the neighbourhood knowledge in X .

3.2. Markov chain

A Markov chain is a stochastic model that describes a sequence of possible events in which the probability of each event depends only on what was attained in the previous event [25]. This chain exists in a discrete exponential set and state space. An irreducible and positive recurrent Markov chain is a strictly stationary Markov chain with a unique stationary distribution vector. The sequence of nodes (or states) visited by a random walker on a specific network forms a Markov chain [10]. This process can be represented by the state transition probability matrix $\mathbf{P}$, a positive definite matrix where $p_{ij}^{(t)}$ represents the probability that the random walker moves from node i to j at time t . The state transition probability of moving from one state to another is dependent only on the current state and remains invariant across the whole process. The state transition process can be represented by

$$p_{ij}^{(t)} = p_{ij}^{(t-1)} = p(x^{(t)} = j | x^{(t-1)} = i) \quad (\forall i \sum_j p_{ij} = 1). \quad (3)$$

A simple example in Fig. 1 is given to illustrate the process of a Markov random walk.

In Fig. 1, there are four nodes in a directed network, p_{12} represents the probability that the random walker moves from node 1 to 2, and $p_{12} + p_{14} = 1$. The random walker starts at node 1 (Fig. 1(a)) and selects the next node according to the state transition probability and the connectivity between nodes. When arriving at node 2, the random walker jumps from node 2 to node 3 as there is only one directed connection starting from node 2. Given a network and a starting point, the random walker jumps to a node according to the state transition probability and repeats the above process with the current node as the starting point. The sequence of selected nodes represents a random walk process on the network.

The long-range trend of a regular Markov chain is known as its equilibrium or stationary distribution, which could be used to describe the visited probability for each object of the random walk process [41]. The stationary distribution vector π for the random walk can be formulated as:

$$\pi^{(t)} = \pi^{(t-1)} \times \mathbf{P}, \quad (4)$$

where $\pi^{(t)}$ is a row vector representing the stable transition probability vector of x_i , and $\mathbf{P}$ denotes the state transition probability matrix of the random walk.

4. Proposed methodology

The proposed methodology and its algorithm will be discussed in detail in this section, and then an example is demonstrated accordingly.

4.1. Theoretical method

This subsection describes the proposed outlier detection method for a DMA, and the method includes three parts (Fig. 2): a) obtaining the neighbourhood information of objects, b) building the WNIN, and c) constructing the index to measure the inlier degree of objects.

Min–max normalization is adopted in this paper to preprocess numerical data. The normalized data, denoted as x_i^h , are defined as:

$$x_i^h = (x_i^{h'} - \min_{i=1,2,\dots,n} \{x_i^{h'}\}) / (\max_{i=1,2,\dots,n} \{x_i^{h'}\} - \min_{i=1,2,\dots,n} \{x_i^{h'}\}) \in (0, 1), \quad (5)$$

where $x_i^{h'}$ denotes the original value of x_i on numerical attribute c^h ($h = 1, 2, \dots, k$). $\max_{i=1,2,\dots,n} \{x_i^{h'}\}$ and $\min_{i=1,2,\dots,n} \{x_i^{h'}\}$ denote the maximum and minimum values of x_i on numerical attribute c^h in X , respectively.

Datasets collected from industrial systems contain mixed-valued attributes (i.e., numerical and categorical attributes). To solve this issue, an improved heterogeneous distance function [39] is employed in this paper to represent the neighbourhood distance function. The improved heterogeneous distance function of $x_i, x_j \in X$ with respect to C is defined as follows:

$$D_{ij} = \sqrt{\sum_{h=1}^m (d_{ij}^h)^2}, \quad (6)$$

where

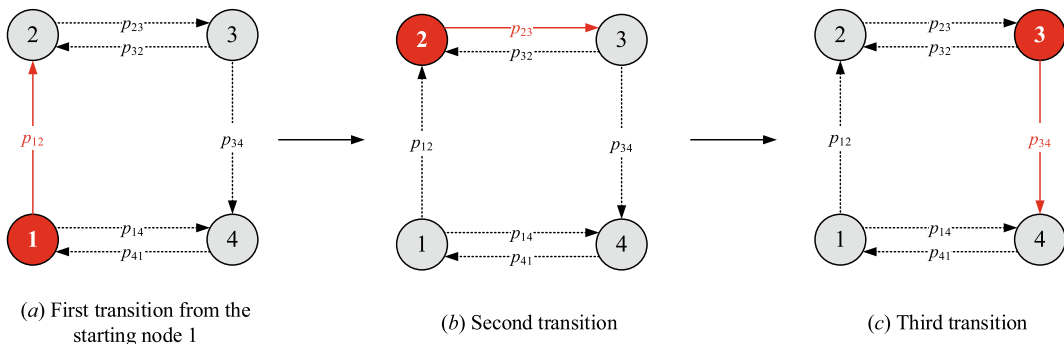


Fig. 1. Markov random walk process.

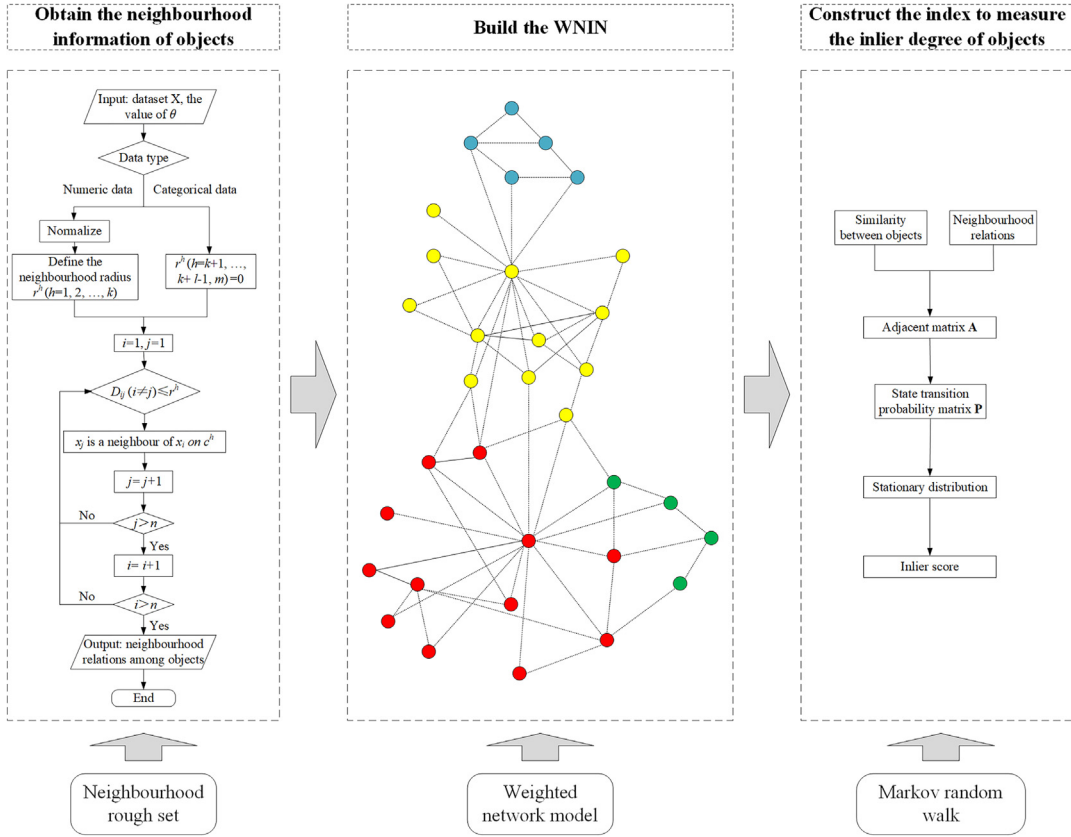


Fig. 2. Framework of the proposed methodology.

$$d_{ij}^h = \begin{cases} 1, & \text{if } x_i \text{ and } x_j \text{ are unknown;} \\ |x_i^h - x_j^h|, & \text{when } h = 1, 2, \dots, k; \\ 0, & x_i^h = x_j^h \quad (h = k+1, k+2, \dots, m); \\ 1, & x_i^h \neq x_j^h \quad (h = k+1, k+2, \dots, m), \end{cases} \quad (7)$$

where D_{ij} represents the total distance between objects x_i and x_j on C , d_{ij}^h represents the distance between object x_i and object x_j on special attribute c^h ($h = 1, 2, \dots, k$), and $|x_i^h - x_j^h| = \sqrt{(x_i^h - x_j^h)^2}$ measures the Euclidean distance between numerical objects x_i and object x_j .

The traditional neighbourhood radius r was defined with some limitations, e.g., fixed values for all attributes are usually assigned according to expert experience [5]. The value of r^h determines the construction of the neighbourhood information network; thus, introducing some objective information of the single attribute c^h would be more adaptive and reasonable. For this purpose, we apply the coefficient of variation and a subjected adjustment parameter to establish the neighbourhood radius r^h so that it is more adaptive and objective for different application scenarios. It is defined as follows:

$$r^h = \begin{cases} \sigma^h / (\theta \mu^h), & \text{when } h = 1, 2, \dots, k; \\ 0, & \text{when } h = k+1, k+2, \dots, m, \end{cases} \quad (8)$$

where σ^h and μ^h represent the standard deviation and the mean value of the objects on numerical attribute c^h , respectively, and θ is a given parameter ($\theta > 0$) for the radius adjustment, which can ensure that the proposed method is adaptive to the neighbourhood radii of different types of distribution of datasets on each numerical attribute.

In the WNIN, the nearest neighbour domain of x_i is not considered to avoid the random walker jumping back and forth to x_i . The neighbour N^h of x_i on c^h is defined as follows:

$$N_i^h = \{x_j (j \neq i) | x_i, x_j \in X, d_{ij}^h \leq r^h\}. \quad (9)$$

Furthermore, the neighbourhood relation R^h of X on c^h is defined as follows:

$$R^h = \{(x_i, x_j) \in X \times X | x_j \in N_i^h\}. \quad (10)$$

It is assumed that the neighbourhood relation-induced neighbourhood knowledge is as follows:

$$X/R^h = \{\varepsilon_1^h, \varepsilon_2^h, \dots, \varepsilon_n^h\} \quad (h = 1, 2, \dots, m), \quad (11)$$

where $\varepsilon_i^h (i = 1, 2, \dots, n)$ represents the number of neighbours for the i^{th} object on c^h .

If and only if object x_j belongs to the neighbour of object x_i is there an edge between x_i and x_j . In addition, the relative weight is the similarity between x_i and x_j on C , denoted as S_{ij} , as follows:

$$S_{ij} = \begin{cases} 0, & \text{if } i \neq j; \\ \sum_{h=1}^m w^h s_{ij}^h, & \text{otherwise,} \end{cases} \quad (12)$$

where w^h is the weight of c^h , and s_{ij}^h is the similarity of x_i and x_j on c^h . s_{ij}^h is defined as:

$$s_{ij}^h = \begin{cases} 1 - |x_i^h - x_j^h|, & \text{when } h = 1, 2, \dots, k; \\ 1, & x_i^h = x_j^h \quad (h = k + 1, k + 2, \dots, m); \\ 0, & x_i^h \neq x_j^h \quad (h = k + 1, k + 2, \dots, m). \end{cases} \quad (13)$$

The weight of c^h is calculated using the entropy weight method [8]. The information entropy E of c^h is defined as

$$E^h = -(\ln m^{-1}) \sum_{i=1}^n (|\varepsilon_i^h| / (\sum_{i=1}^n |\varepsilon_i^h|)) (\ln |\varepsilon_i^h| / (\sum_{i=1}^n |\varepsilon_i^h|)) \quad (h = 1, 2, \dots, m), \quad (14)$$

where $|\varepsilon_i^h|$ represents the cardinality of the set ε_i^h . Then, the weight of c^h can be obtained using the following:

$$w^h = (1 - E^h) / (m - \sum_{h=1}^m E^h). \quad (15)$$

The neighbours of each object are used to generate the edges on the WNIN. The adjacent matrix (**A**), which involves the nonzero element that is the weight describing the similarity between the nodes of the corresponding edge, can be defined as the product of two square matrices **S** and **U**.

$$\mathbf{A} = \mathbf{S} \circ \mathbf{U}, \quad (16)$$

where the operator ‘ $\circ$ ’ denotes the Hadamard product, which is equivalent to the elementwise product of the two matrices with equal dimensions [14]. **S** is a square matrix whose elements are S_{ij} ($i, j = 1, 2, \dots, n$). Matrix **U** is a binary matrix containing the neighbourhood information of each node. The elements in matrix **U** are defined as follows:

$$U_{ij} = \begin{cases} 1, & \text{if } x_j \in N_i^h \quad (h = 1, 2, \dots, m); \\ 0, & \text{otherwise,} \end{cases} \quad (17)$$

practically, $U_{ij} = 1$ can be altered by another condition according to the actual case.

The state transition probability matrix **P** can be calculated by normalizing the adjacent matrix **A** by rows:

$$\mathbf{P} = \mathbf{A} \times \mathbf{B}^{-1}, \quad (18)$$

where **B** is a diagonal matrix of **A** in which each element in the diagonal matrix equals the sum of the corresponding row elements, and **P** is a Markov random walk matrix. The sum of the jump probability of each point and all the other nodes is 1. Starting from any state at any time, the process stabilises after a certain number of iterations. The stationary distribution vector can be defined as follows:

$$\boldsymbol{\pi}^{(t)} = \boldsymbol{\pi}^{(t-1)} \times \mathbf{P}, \quad (19)$$

where $\boldsymbol{\pi}^{(t)} = (\pi_1^{(t)}, \pi_2^{(t)}, \dots, \pi_n^{(t)})$, and $\pi_i^{(t)}$ ($i = 1, 2, \dots, n$) represents the probability of a random walker remaining at node x_i at time step t . Specifically, when $t = 0$, $\pi_i^{(0)}$ ($i = 1, 2, \dots, n$) = $1/n$. This means that the initial importance of x_i is $\pi^{(0)} = (1/n, 1/n, \dots, 1/n)$.

Given that the initial time step t is at node x_i , we want the random walker to jump to the nearest node x_j at $t + 1$. The transition probability of our customised Markov random walk process is calculated using the similarity between the nodes, and this approach can effectively complete the transition process between the nodes. The nodes that tend to be outliers (isolated nodes) are less likely to be accessed. After the Markov random walk reaches the equilibrium, each element in the stationary distribution vector can represent the probability of the walker remaining on each node (the probability that the corresponding node is accessed), i.e., the importance of each node. A larger stationary distribution vector value of a node

corresponds to a smaller probability of the trend outlier. However, this value is necessary but not sufficient for outlier detection. Therefore, another essential element is worth introducing to achieve better effectiveness.

The values in the row vector π are all decimal values and are nearly similar, and it is not easy to distinguish the importance of the nodes. To obtain a higher importance score for each node, denoted as φ_i , we standardise the values as follows:

$$\varphi_i = (b - a)(\pi_i^{(t)} - \min\{\pi^{(t)}\}) / (\max\{\pi^{(t)}\} - \min\{\pi^{(t)}\}) + a \quad (i = 1, 2, \dots, n), \quad (20)$$

where $a, b \in R^+$, which satisfy $a < b$; and $\min\{\pi^{(t)}\}$ and $\max\{\pi^{(t)}\}$ represent the minimum and maximum values in $\pi^{(t)}$, respectively. Eq. (20) scales the importance scores of all the nodes to the interval $[a, b]$. In this paper, a and b are set as 1 and 100, respectively.

The importance of a node is closely related to its number of neighbours. The value of the adjustment parameter θ for the neighbourhood radius r has a considerable influence on the out-degree of nodes. Therefore, a new index, i.e., the inlier score, denoted as ξ_i , is introduced to solve the above issues. It is defined as:

$$\xi_i = \varphi_i / \sum_{j=1}^n U_{ij} \quad (i = 1, 2, \dots, n). \quad (21)$$

Considering the previous principle that the potential outlier objects should be assigned less importance, the score allocated to a real node in the WNIN should be inversely proportional to its potential outliers. In this manner, an outlier object is assigned a relatively smaller inlier score than the inliers.

The convergence of the Markov random walk process based on the WNIN is proven in [Appendix A](#).

4.2. Algorithm

The algorithm for the proposed outlier detection methodology is presented in Algorithm 1.

Algorithm 1: WNIN Based Outlier Detection (WNINOD) Algorithm

Input: Dataset $X \in R_{n \times m}^+$, θ is a parameter for the radius adjustment.
Output: Inlier score of each data object.

```

1 for  $i = 1 : n$  do
2   for  $h = 1 : k$  do
3      $x_i^h \leftarrow (x_i^{h'} - \min_{l=1,2,\dots,n} x_l^{h'}) / (\max_{l=1,2,\dots,n} x_l^{h'} - \min_{l=1,2,\dots,n} x_l^{h'}) \in (0, 1)$ 
4      $N_i^h \leftarrow \{x_j (j \neq i) \mid x_i, x_j \in X, d_{ij}^h \leq r^h\}$ 
5   end
6 end
7 for  $h = 1 : m$  do
8   for  $i = 1 : n$  do
9      $R^h \leftarrow \{(x_i, x_j) \in X \times X \mid x_j \in N_i^h\}$ 
10     $W^h \leftarrow \{(1 - E^h) / (m - \sum_{h=1}^m E^h)\}$ 
11  end
12 end
13 for  $i = 1 : n$  do
14   for  $j = 1 : n$  do
15      $S_{ij} \leftarrow \begin{cases} 0, & \text{if } i \neq j; \\ \sum_{h=1}^m w^h s_{ij}^h, & \text{otherwise,} \end{cases}$ 
16      $A_{ij} \leftarrow S_{ij} U_{ij}$ 
17   end
18 end
19  $P \leftarrow A \times B^{-1}$ 
20  $\pi^{(0)} = (1/n, 1/n, \dots, 1/n)$ 
21 while not converged do
22    $\pi^{(t)} = \pi^{(t-1)} \times P$ 
23 end
24 for  $i = 1 : n$  do
25    $\xi_i = \varphi_i / \sum_{j=1}^n U_{ij}$ 
26 end
```

4.3. An illustrative example

An illustrative example is presented to demonstrate the proposed theoretical method.

Example 1. An neighbourhood information system $NIS = (X, C, V, f)$ is described in Table 1, where $X = \{x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9, x_{10}\}$ and $C = \{c^1, c^2, c^3, c^4\}$.

According to Eq. (5), the standardized data is considered and the relative standard deviation and mean of the numeric data are calculated, as shown in Table 2. Next, considering $\theta = 2$, the self-adaptive radius r^h is determined using Eq. (8), and the result is presented in the last row of Table 2.

Next, the neighbours of each object on C are determined using Eqs. (9)–(11) and listed in Table 3.

The unweighted neighbourhood information network for Example 1 is drawn, as shown in Fig. 3.

The state transition probability matrix $\mathbf{P}$ can be calculated using Eqs. (12)–(18).

$$\mathbf{P} = \begin{bmatrix} 0 & 0.1024 & 0.1271 & 0.1299 & 0.2148 & 0.1636 & 0.1103 & 0 & 0.1519 & 0 \\ 0.0764 & 0 & 0.0997 & 0.1922 & 0.0997 & 0.0677 & 0.0892 & 0.1207 & 0.164 & 0.0903 \\ 0.1116 & 0.1173 & 0 & 0.1185 & 0.0956 & 0.1013 & 0.2355 & 0 & 0.1002 & 0.1199 \\ 0.0936 & 0.1857 & 0.0973 & 0 & 0.0765 & 0.0852 & 0.0778 & 0.1363 & 0.1782 & 0.0694 \\ 0.231 & 0.1437 & 0.1172 & 0.1142 & 0 & 0.0894 & 0.1021 & 0 & 0.0989 & 0.1036 \\ 0.1557 & 0.0864 & 0.1099 & 0.1125 & 0.0791 & 0 & 0.1037 & 0 & 0.1459 & 0.2069 \\ 0.1064 & 0.1153 & 0.2587 & 0.1041 & 0.0915 & 0.1051 & 0 & 0.0611 & 0 & 0.1577 \\ 0 & 0.238 & 0 & 0.2783 & 0 & 0 & 0.0933 & 0 & 0.2992 & 0.0912 \\ 0.1286 & 0.186 & 0.0966 & 0.2092 & 0.0778 & 0.1297 & 0 & 0.1721 & 0 & 0 \\ 0 & 0.1355 & 0.1529 & 0.1078 & 0.1078 & 0.2434 & 0.1832 & 0.0694 & 0 & 0 \end{bmatrix}$$

After a certain number of iterations, the stationary distribution vector is obtained as follows:

$$\pi^{(t)} = [0.0936, 0.1257, 0.1067, 0.1301, 0.0872, 0.0986, 0.0974, 0.0637, 0.1109, 0.0836]$$

Finally, the inlier score of each object can be calculated using Eqs. (20) and (21),

$$\begin{aligned} \xi_1 &= \varphi_1 / \sum_{j=1}^{10} N_{1j} = 45.5798/7 = 6.5114 & \xi_2 &= 10.3822 & \xi_3 &= 8.1389 & \xi_4 &= 11.1111 \\ \xi_5 &= 4.5047 & \xi_6 &= 6.6293 & \xi_7 &= 6.4057 & \xi_8 &= 0.2 & \xi_9 &= 10.1962 & \xi_{10} &= 4.3815 \end{aligned}$$

The inlier scores of these objects are shown in Fig. 4.

As shown in Fig. 4, x_8 has the minimum inlier score (0.2); therefore, it is more likely to be an outlier according to the proposed methodology.

5. Experimental analysis

The UCI datasets are used to verify the effectiveness of the proposed method in this section. We evaluate our theory and the proposed algorithms by using a PC with a 2.40 GHz Intel Core i7-5500U CPU and 8.00 GB of memory operating Windows 10. The algorithms are programmed using Visual Studio 2013.

Two datasets from the UCI repository [2] are used to verify the effectiveness and feasibility of the proposed outlier detection method considering the change in the radius adjustment parameter θ and four existing methods. Five benchmark methods are analysed in Table 4. To enhance the comparability of the experimental results for each outlier detection method, the

Table 1
Original information system for Example 1.

X	c^1	c^2	c^3	c^4
x_1	12	0.4	A	q
x_2	5	0.2	B	m
x_3	8	0.3	D	h
x_4	10	0.1	B	m
x_5	2	0.1	A	q
x_6	13	0.5	C	q
x_7	7	0.7	D	h
x_8	20	0.9	B	m
x_9	14	0.2	B	q
x_{10}	6	0.8	C	h

Table 2
Standardized information system for [Example 1](#).

X	c^1	c^2	c^3	c^4
x_1	5/9	0.375	A	q
x_2	1/6	0.125	B	m
x_3	3/9	0.25	D	h
x_4	4/9	0	B	m
x_5	0	0	A	q
x_6	11/18	0.5	C	q
x_7	5/18	0.75	D	h
x_8	1	1	B	m
x_9	6/9	0.125	B	q
x_{10}	2/9	0.875	C	h
σ^h	0.2756	0.3482	N/A	N/A
μ^h	0.4278	0.4	N/A	N/A
r^h	0.322	0.43555	0	0

Table 3
Neighbours of each object on C .

X	c^1	c^2	c^3	c^4
x_1	$\{x_3, x_4, x_6, x_7, x_9\}$	$\{x_2, x_3, x_4, x_5, x_6, x_7, x_9\}$	$\{x_5\}$	$\{x_5, x_6, x_9\}$
x_2	$\{x_3, x_4, x_7, x_9, x_9\}$	$\{x_1, x_3, x_4, x_5, x_6, x_9\}$	$\{x_4, x_8, x_9\}$	$\{x_4, x_8\}$
x_3	$\{x_1, x_2, x_4, x_6, x_7, x_{10}\}$	$\{x_1, x_2, x_4, x_5, x_6, x_9\}$	$\{x_7\}$	$\{x_7, x_{10}\}$
x_4	$\{x_1, x_2, x_3, x_6, x_7, x_9, x_{10}\}$	$\{x_1, x_2, x_3, x_5, x_9\}$	$\{x_2, x_8, x_9\}$	$\{x_2, x_8\}$
x_5	$\{x_2, x_7, x_{10}\}$	$\{x_1, x_2, x_4, x_5, x_9\}$	$\{x_1\}$	$\{x_1, x_6, x_9\}$
x_6	$\{x_1, x_3, x_4, x_9\}$	$\{x_1, x_2, x_3, x_6, x_7, x_9\}$	$\{x_{10}\}$	$\{x_1, x_5, x_9\}$
x_7	$\{x_1, x_2, x_3, x_4, x_5, x_{10}\}$	$\{x_1, x_6, x_7, x_9\}$	$\{x_3\}$	$\{x_3, x_{10}\}$
x_8	N/A	$\{x_7, x_9\}$	$\{x_2, x_4, x_9\}$	$\{x_2, x_4\}$
x_9	$\{x_1, x_4, x_6\}$	$\{x_1, x_2, x_3, x_4, x_5, x_6\}$	$\{x_2, x_4, x_8\}$	$\{x_1, x_5, x_6\}$
x_{10}	$\{x_2, x_3, x_4, x_5, x_7\}$	$\{x_6, x_7, x_9\}$	$\{x_6\}$	$\{x_3, x_7\}$

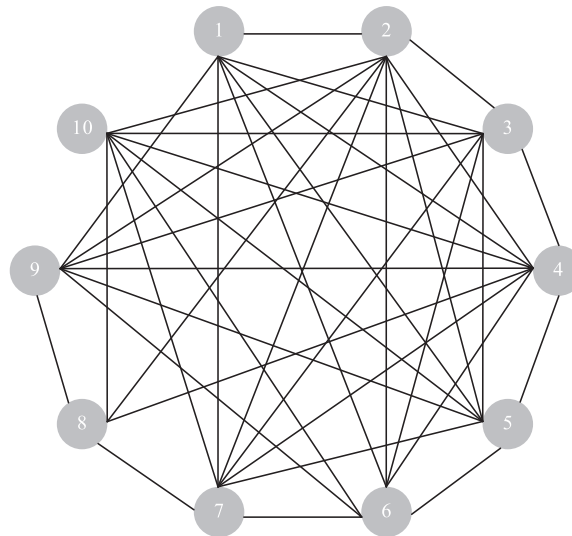


Fig. 3. Unweighted neighbourhood information network for [Example 1](#).

evaluation index system proposed in [1] is adopted. An outlier detection approach can be evaluated by considering the identification ratio between the practical and detected outliers. A larger ratio of the actual outliers corresponds to a higher effectiveness and performance of the method.

5.1. Hayes-Roth dataset

The Hayes-Roth dataset is used in the human subject transfer tests. It contains 132 objects, 5 numeric-valued condition attributes in which the actual values of the attributes (i.e., hobby has the values chess, sports and stamps) are replaced with

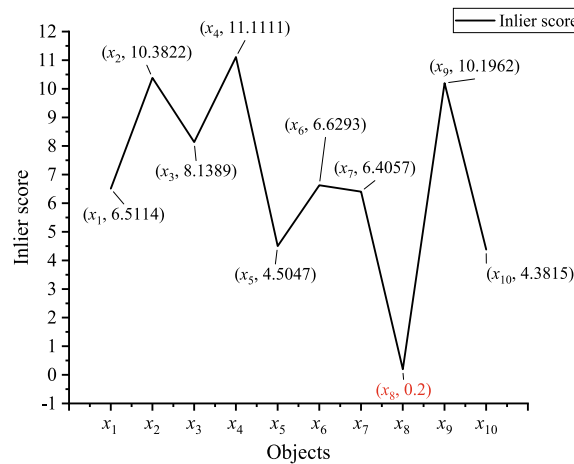


Fig. 4. Line chart for the inlier scores of objects.

numeric values, and 1 decision attribute. The objects are classified into 3 categories with 51, 51, and 30 objects in each category, respectively. Class 3 includes the fewest objects, and it can be treated as the rare class. The specific class distribution is presented in Fig. 5.

The proposed method involves only one parameter θ , which is the neighbourhood radius adjustment parameter (Eq. (9)); and the detection results are affected by the fluctuations of θ . Thus, we discuss the sensitivity of θ before comparing the proposed approach with other methods. For the given dataset, the range of the neighbourhood radius depends on the variation coefficient of the value of the numerical attribute. A smaller θ results in a larger neighbourhood radius, which leads to fewer outliers; however, a larger θ results in a smaller neighbourhood radius, which leads to more outliers. Min-max normalization makes the numerical data fall in $[0, 1]$, which leads to the maximum distance between objects being 1. Therefore,

$\min_{i \neq j; i, j=1, 2, \dots, n} \{d_{ij}^h\} \leq r^h \leq 1$. It is calculated that $0.99 \leq \theta \leq 76.50$. The step-by-step calculation is adopted. First, let θ be $[1, 25]$ and the step size is 1 to reduce the operating time. The preliminary calculation results for the changes in θ are shown in Table 5.

The acquisition details and interpretation of the results in Table 5 are as follows:

- (1) For the proposed WNINOD, the inlier scores of all the objects are calculated and sorted in ascending order, which corresponds to an object sequence. According to the ascending order of the inlier scores, “Top ratio $n\%$ ” and “Number of objects” represent the proportion n of the front objects that are selected and the number of selected objects in the Hayes-Roth dataset, respectively.
- (2) Furthermore, the “Number of rare classes included” denotes the number of identified objects that are actual outliers, and “Coverage” denotes the ratio of the “Number of rare classes included” to the total number of actual outliers in the Hayes-Roth dataset. Therefore, the “Top ratio (Number of objects)” and “Number of rare classes included (Coverage)” can act as the premise and result, respectively, to establish a systematic and dynamic estimation of the identification accuracy.

Table 4

Introduction of the five outlier detection methods.

Name [Reference]	Strategy	Strengths//Weaknesses
VOS [36]	Random walk	Exploits the local information from a global perspective.//The similarity between the objects is related to the number of data objects in the subset. The calculation of the pairwise similarities and stationary distribution vector is time-consuming. The method is not suitable for categorical data and mixed-valued data.
NED [5]	Rough sets	Adaptable for both categorical and numeric data.//Complex neighbourhood calculation and subjective radius selection.
OutRank [24]	Markov model	Simple calculation process.//Limited by the cosine similarity measure. The method is not suitable for categorical data and mixed-valued data.
NIEOD [42]	Neighbourhood information entropy	Adaptable for categorical, numeric, and mixed-valued data.// Information fusion of only single attributes.
WNINOD	Markov random walk, neighbourhood rough set	Adaptable for numeric, categorical, and mixed-valued data. Exhibits strong interpretability.//The calculation of the stationary distribution vector is time-consuming.

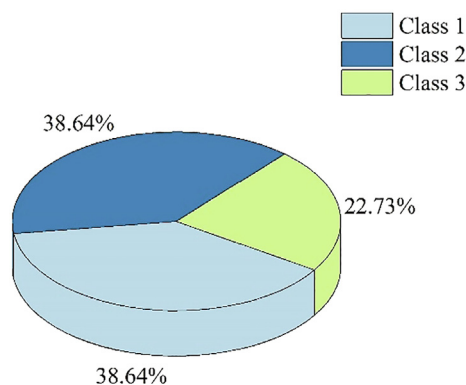


Fig. 5. Classification of the Hayes-Roth dataset.

(3) Moreover, “Mean” denotes the average “Number of rare classes included” in the detection results with a different “Number of objects”. It can be used to measure the outlier detection accuracy of the proposed method as θ changes, and the detection accuracy rate is directly proportional to the “Mean”.

Table 5

Preliminary calculation results for the changes in θ in the Hayes-Roth dataset.

Top ratio (Number of objects)	Number of rare classes included (Coverage (%))						
	$\theta = 1$	$\theta = 2$	$\theta = 3$	$\theta = 4$	$\theta = 5$	$\theta = 6$	$\theta = 7$
7%	9	9	9	9	9	9	9
(9)	(30)	(30)	(30)	(30)	(30)	(30)	(30)
13%	15	17	17	17	17	17	17
(17)	(50)	(57)	(57)	(57)	(57)	(57)	(57)
16%	17	21	21	21	21	21	21
(21)	(57)	(70)	(70)	(70)	(70)	(70)	(70)
19%	20	24	24	25	25	25	25
(25)	(67)	(80)	(80)	(83)	(83)	(83)	(83)
22%	21	28	27	27	27	27	27
(29)	(70)	(93)	(90)	(90)	(90)	90	
25%	21	30	28	28	28	28	28
(33)	(70)	(100)	(93)	(93)	(93)	(93)	(93)
27%	29	30	29	30	30	30	29
(35)	(97)	(100)	(97)	(100)	(100)	(100)	(97)
38%	25	30	30	30	30	30	30
(50)	(83)	(100)	(100)	(100)	(100)	(100)	(100)
Mean	18.625	23.625	23.125	23.375	23.375	23.375	23.25
Running time (s)	6.17	901.50	780.38	9.57	14.24	465.39	5.42

Top ratio (Number of objects)	Number of rare classes included (Coverage (%))						
	$\theta = 8$	$\theta = 9$	$\theta = 10$	$\theta = 11 - 12$	$\theta = 13 - 15$	$\theta = 16 - 19$	$\theta = 20 - 25$
7%	9	9	9	9	9	9	9
(9)	(30)	(30)	(30)	(30)	(30)	(30)	(30)
13%	17	17	17	17	17	17	17
(17)	(57)	(57)	(57)	(57)	(57)	(57)	(57)
16%	21	21	21	21	21	21	21
(21)	(70)	(70)	(70)	(70)	(70)	(70)	(70)
19%	25	25	25	25	25	25	25
(25)	(83)	(83)	(83)	(83)	(83)	(83)	(83)
22%	27	27	27	27	27	27	27
(29)	(90)	(90)	(90)	(90)	(90)	(90)	(90)
25%	28	28	28	28	28	27	27
(33)	(93)	(93)	(93)	(93)	(93)	(90)	(90)
27%	29	29	29	29	29	29	29
(35)	(97)	(97)	(97)	(97)	(97)	(97)	(97)
38%	30	30	30	30	30	30	30
(50)	(100)	(100)	(100)	(100)	(100)	(100)	(100)
Mean	18.625	23.625	23.125	23.375	23.375	23.375	23.25
Running time (s)	5.42	5.42	5.42	5.42	5.42	1492.97	1346.38

(4) As shown in Table 5, the WNINOD with $\theta = 2$ has a better result than those of the other conditions. For example, when the top ratio is 25% (the number of objects is 33), the WNINOD with $\theta = 2$ identifies 30 (coverage 100%) actual outliers. In contrast, the numbers of actual outliers identified are 21 (coverage 70%) with $\theta = 1$, 28 (coverage 93%) with $\theta = 13 - 15$, and 27 (coverage 90%) with $\theta = 16 - 25$.

For better visualization, the preliminary calculation results for the changes in θ in Table 5 are illustrated in Fig. 6.

Fig. 6 shows the outlier detection accuracy superiority of the WNINOD as θ changes. When θ is in $[1, 2)$, the “Mean” gradually increases (i.e., the outlier detection accuracy gradually increases) and reaches the highest value when $\theta = 2$. Meanwhile, the neighbourhood radius decreases as θ increases, and more objects are considered outliers, which further leads to a decrease in the accuracy of the recognition results. Therefore, the optimal value is near 2. Next, we set θ equal to 1.5 and 2.5, respectively, and the optimal solution is near 2.5 after the calculation. Finally, we set θ in $(2, 3)$ and the step size as 0.1 (the step size can be set according to the actual needs). The detection results are shown in Table 6.

When θ is 2.4, 2.5, 2.6, 2.7, 2.8 and 2.9, the detection accuracy reaches the optimum. Moreover, the relative running times are 9.30 s, 478.34 s, 16.50 s, 3.31 s, 9.75 s, and 15.11 s, respectively. In summary, the detection accuracy and efficiency can reach the optimum when $\theta = 2.7$.

Here, we compare the detection results of the proposed method with those of four other outlier detection methods (as shown in Fig. 7 and Table 7). Due to the different outlier measurement mechanisms, the result analysis should strictly follow the mechanism of the related method. For the WNINOD and OutRank, the scores of all the objects are calculated and sorted in ascending order, which naturally corresponds to an object sequence; while the calculation results of the VOS, NED, and NIEOD are sorted in descending order.

As shown in Table 7, when the top ratio is 25% (the total number of objects is 33), the WNINOD identifies all the actual outliers (30 (coverage 100)) in the Hayes-Roth dataset while the numbers of actual outliers identified using the VOS, the NED, OutRank, and the NIEOD are 8 (coverage 27%), 29 (coverage 97%), 9 (coverage 30%), and 8 (coverage 27%), respectively. For all ten levels of the top ratio (number of objects), the WNINOD exhibits the best performance compared with those of the other four methods.

For better visualisation, the experimental results (Table 7) are also illustrated in Fig. 8. The figure clearly reflects the mainstream accuracy superiority of the WNINOD compared to those of the other four methods. Moreover, the computational times for the five approaches (WNINOD, VOS, NED, OutRank, and NIEOD) are 3.31 s, 306.4 s, 0.01 s, 1.5 s, and 1.7 s, respectively. Thus, the time consumed by the WNINOD is relatively higher; however, this time cost is practically acceptable to obtain a better identification result.

In summary, the WNINOD exhibits better effectiveness and performance for the Hayes-Roth dataset, which has only numerical attributes, compared to those of the other four approaches (VOS, NED, OutRank, and NIEOD).

5.2. Lymphography dataset

The Lymphography dataset is one of three domains provided by the Oncology Institute that has repeatedly been used in machine learning research. It contains 148 objects, 18 condition attributes (i.e., 3 numeric attributes and 15 categorical attributes), and 1 decision attribute. The 148 objects are classified into 4 categories (normal find, metastases, malign lymph, and fibrosis) (as shown in Fig. 9), which included 2, 81, 61 and 4 objects in each category, respectively. The “normal find” and “fibrosis” classes are regarded as rare classes with a total of 6 actual outliers.

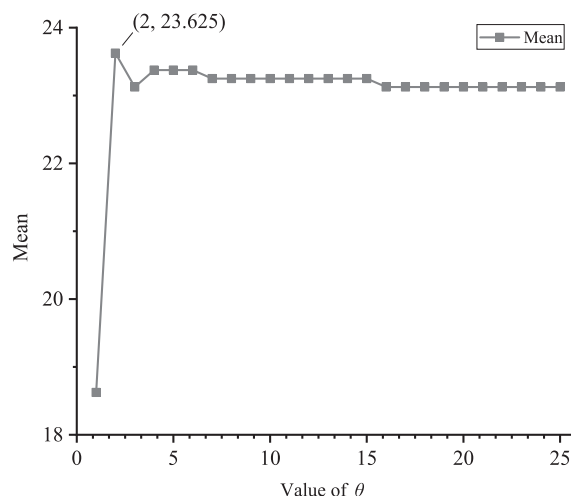
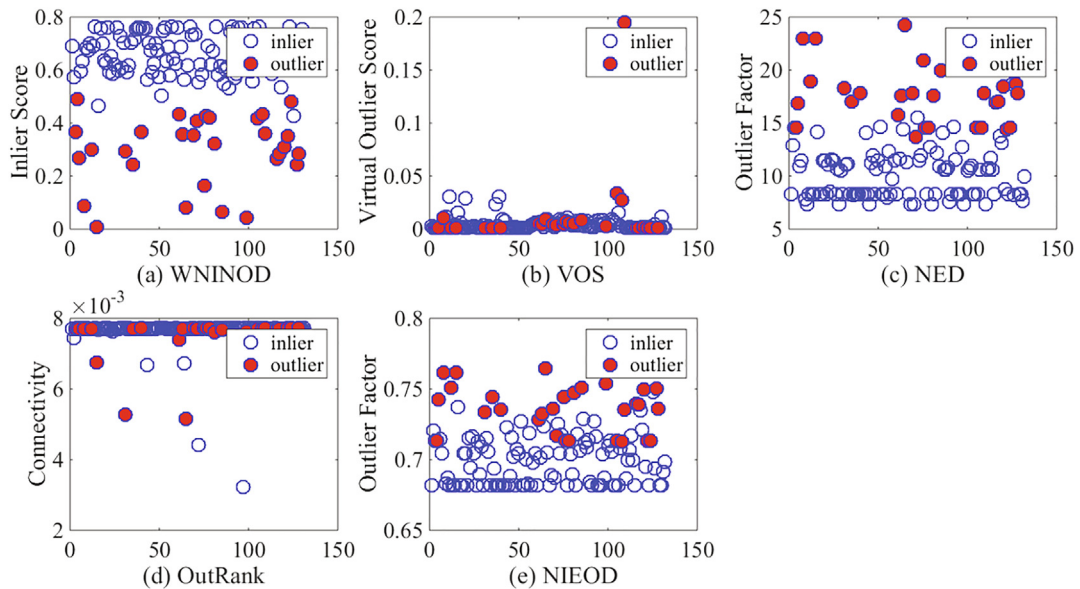


Fig. 6. Sensitivity analysis for the changes of θ in the Hayes-Roth dataset.

Table 6Selection of the optimal value of θ in the Hayes-Roth dataset.

Top ratio (Number of objects)	Number of rare classes included (Coverage (%))								
	$\theta = 2.1$	$\theta = 2.2$	$\theta = 2.3$	$\theta = 2.4$	$\theta = 2.5$	$\theta = 2.6$	$\theta = 2.7$	$\theta = 2.8$	$\theta = 2.9$
7%	9	9	9	9	9	9	9	9	9
(9)	(30)	(30)	(30)	(30)	(30)	(30)	(30)	(30)	(30)
13%	17	17	17	17	17	17	17	17	17
(17)	(57)	(57)	(57)	(57)	(57)	(57)	(57)	(57)	(57)
16%	21	21	21	21	21	21	21	21	21
(21)	(70)	(70)	(70)	(70)	(70)	(70)	(70)	(70)	(70)
19%	24	24	24	25	25	25	25	25	25
(25)	(80)	(80)	(80)	(83)	(83)	(83)	(83)	(83)	(83)
22%	28	28	28	28	28	28	28	28	28
(29)	(93)	(93)	(93)	(93)	(93)	(93)	(93)	(93)	(93)
25%	30	30	30	30	30	30	30	30	30
(33)	(100)	(100)	(100)	(100)	(100)	(100)	(100)	(100)	(100)
27%	30	30	30	30	30	30	30	30	30
(35)	(100)	(100)	(100)	(100)	(100)	(100)	(100)	(100)	(100)
38%	30	30	30	30	30	30	30	30	30
(50)	(100)	(100)	(100)	(100)	(100)	(100)	(100)	(100)	(100)
Mean	23.625	23.625	23.625	23.75	23.75	23.75	23.75	23.75	23.75
Running time (s)	917.37	6.99	699.77	9.30	478.34	16.50	3.31	9.75	15.11

**Fig. 7.** Experimental results of the five outlier detection methods for the Hayes-Roth dataset.**Table 7**

Experimental analysis of the five outlier detection methods for the Hayes-Roth dataset.

Top ratio (Number of objects)	Number of rare classes included (Coverage (%))				
	WNINOD	VOS	NED	OutRank	NIEOD
7% (9)	9 (30)	4 (13)	8 (27)	5 (17)	4 (13)
13% (17)	17 (57)	6 (20)	16 (53)	8 (27)	7 (23)
16% (21)	21 (70)	6 (20)	20 (67)	9 (30)	7 (23)
19% (25)	25 (83)	8 (27)	21 (67)	9 (30)	7 (23)
22% (29)	28 (93)	8 (27)	25 (83)	9 (30)	7 (23)
25% (33)	30 (100)	8 (27)	29 (97)	9 (30)	8 (27)
27% (35)	30 (100)	9 (30)	29 (97)	9 (30)	10 (33)
38% (50)	30 (100)	14 (47)	30 (100)	10 (33)	13 (43)
50% (66)	30 (100)	15 (50)	30 (100)	11 (37)	17 (57)
80% (105)	30 (100)	22 (73)	30 (100)	23 (77)	26 (87)
Running time (s)	3.31	306.4	0.01	1.5	1.7

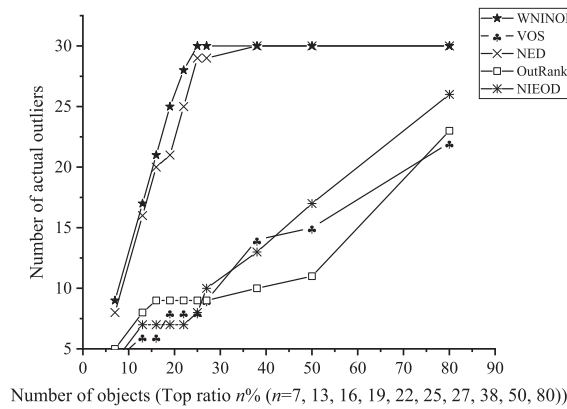


Fig. 8. Comparison of the experimental results for the Hayes-Roth dataset.

When the values of θ are 0.2, 0.5 and 1 (the process of assigning parameter θ is the same as in SubSection 5.1), the detection accuracy reaches the optimal value; and the running times are 31.79 s, 30.67 s and 18.6 s, respectively. In summary, the WNINOD with $\theta = 1.0$ is better than the other WNINODs in terms of the top ratio (number of objects) and running time.

The outlier detection results and comparative analyses for the five methods (WNINOD, VOS, NED, OutRank, and NIEOD) are provided in Fig. 10 and Table 8. For better visualisation, the experimental results (i.e., Table 8) are also illustrated in Fig. 11. The WNINOD outperforms the other three algorithms (VOS, NED, and OutRank) in terms of the ten “Top ratio (Number of objects)” levels. For example, when the top ratio is 8.2% (the number of objects is 12), the WNINOD identifies 6 (coverage 100%) actual outliers while the numbers of actual outliers identified using the VOS, the NED, and OutRank are 1 (coverage 16.7%), 4 (coverage 66.7%), and 4 (coverage 66.7%), respectively. Although the identification effects of the WNINOD in the first three levels (0.7% (the number of objects is 1), 3.4% (the number of objects is 5), and 7.5% (the number of objects is 11)) are slightly worse than those of the NIEOD, both methods exhibit the same performance in the last four levels. Moreover, the mentioned five approaches consume 18.6 s, 302.0 s, 17.2 s, 5.1 s, and 4.7 s, respectively; all the methods except for the VOS are reasonably fast.

In summary, although the detection results of the WNINOD are slightly worse than those of the NIEOD, the WNINOD exhibits higher performance for the Lymphography dataset, which includes both numeric and categorical attributes, when compared to the performance of three other methods, namely, the VOS, the NED, and OutRank.

5.3. Summary

Through the above two experiments with the Hayes-Roth and Lymphography datasets, the utility and effectiveness of the proposed approach is summarized as follows:

- (1) The proposed outlier detection method based on the WNINOD can be used to handle datasets with a single type of attribute or mixed-valued attributes.

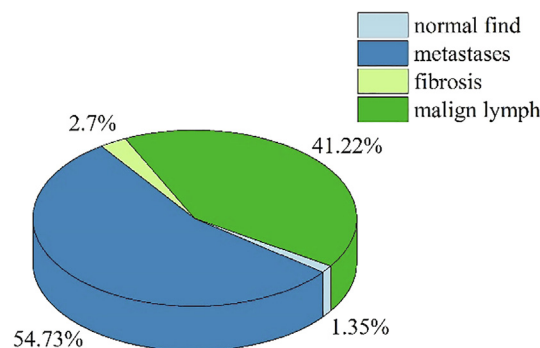


Fig. 9. Classification of the Lymphography dataset.

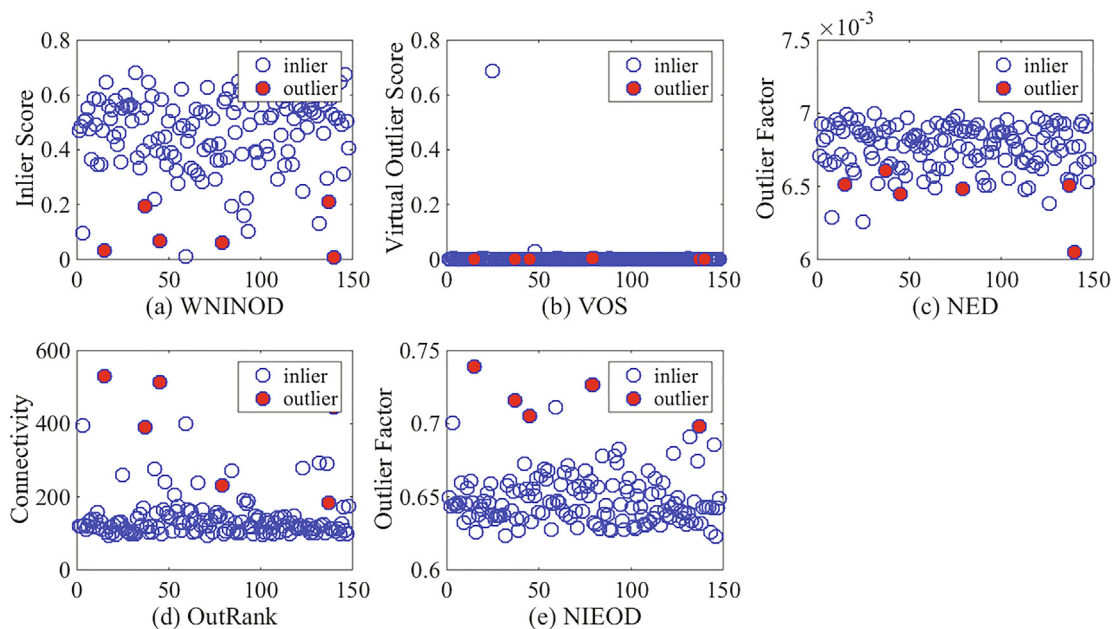


Fig. 10. Experimental results of the five outlier detection methods for the Lymphography dataset.

Table 8

Experimental analysis of the five outlier detection methods for the Lymphography dataset.

Top ratio (Number of objects)	Number of rare classes included (Coverage (%))				
	WNINOD	VOS	NED	OutRank	NIEOD
0.7% (1)	1 (16.7)	0 (0)	1 (16.7)	1 (16.7)	1 (16.7)
3.4% (5)	4 (66.7)	0 (0)	3 (50)	2 (33.3)	5 (83.3)
7.5% (11)	5 (83.3)	1 (16.7)	4 (66.7)	4 (66.7)	6 (100)
8.2% (12)	6 (100)	1 (16.7)	4 (66.7)	4 (66.7)	6 (100)
10.2% (15)	6 (100)	1 (16.7)	5 (83.3)	5 (83.3)	6 (100)
20.3% (30)	6 (100)	2 (33.3)	6 (100)	6 (100)	6 (100)
27.1% (40)	6 (100)	2 (33.3)	6 (100)	6 (100)	6 (100)
33.8% (50)	6 (100)	3 (50)	6 (100)	6 (100)	6 (100)
40.6% (60)	6 (100)	4 (66.7)	6 (100)	6 (100)	6 (100)
77.1% (115)	6 (100)	6 (100)	6 (100)	6 (100)	6 (100)
Running time (s)	18.6	302.0	17.2	5.1	4.7

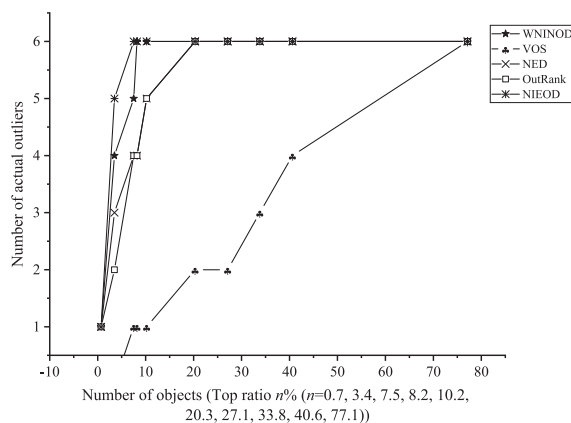


Fig. 11. Comparison of the experimental results for the Lymphography dataset.

- (2) For a given dataset, the range of neighbourhood radius r can be obtained by solving the coefficient of variation and the distance between objects on $c^h (h = 1, 2, \dots, m)$. Thus, the optimal value of θ that can promote the efficiency and accuracy of the proposed method can be optimized according to Eq. (8).
- (3) The improved heterogeneous distance function reflects the degree of closeness of objects in different attribute spaces, and it achieves higher detection accuracy. Besides, the similarity measurement between objects imitates the neighbourhood distance function, which will improve the computational efficiency.
- (4) In addition, the proposed method exhibits excellent interpretability based on the weighted network model, as shown in Fig. 12.

The WNINs are drawn according to the adjacency matrix (**A**) for the Hayes-Roth dataset and Lymphography dataset, respectively. The colour of an edge changes from red to blue as the edge weight increases. The node size is determined by the node importance. The higher the node importance is, the larger the node radius and the more normal the node. In this mechanism, the potential outliers (such as node 15 in Fig. 12(a) and node 140 in Fig. 12(b)) and the relationships among nodes can be easily and clearly interpreted.

6. A real case study

This section is motivated by an engineering requirement for the warehousing management of a company. The warehouse for this electronic tool manufacturing enterprise produces wheat cutters and provides raw materials for 44 production lines, which include preparation, receiving and unloading, tallying, and handling. In the warehousing process, many outlier events (such as damaged labels, unscannable codes, quantities delivered not matching documents, and nonstandardized goods) may occur. The occurrence of these problems has largely reduced the production efficiency and economic benefits of the enterprise. It is very important to monitor the healthy operation of the warehousing system to formulate corresponding measures in time. In [20], the operating status of a warehousing system is described by mixed-valued attributes; 5 numerical attributes and 1 categorical attribute are shown in Table 9.

Li et al. [20] divided the historical operating data of the storage system into 10 groups to adjust the parameters (the radius adjustment parameter λ and the judgement criterion μ). Here, a comparative analysis using the Neighbourhood Information Entropy (NIE) is conducted based on one group randomly. The dataset contains 6 attributes and 70 objects, and five outliers are known. The detection results and comparison analysis based on the WNINOD with the optimized $\theta = 1.6$ and the NIE are shown in Fig. 13 and Table 10, respectively.

From Fig. 13 and Table 10, several conclusions are drawn as follows.

- (1) The “Mean” denotes the average number of actual outliers identified when the top ratio is 34.3%. The values of “Mean” for the two methods are the same, which indicates that they have similar adaptability to this issue to some extent.

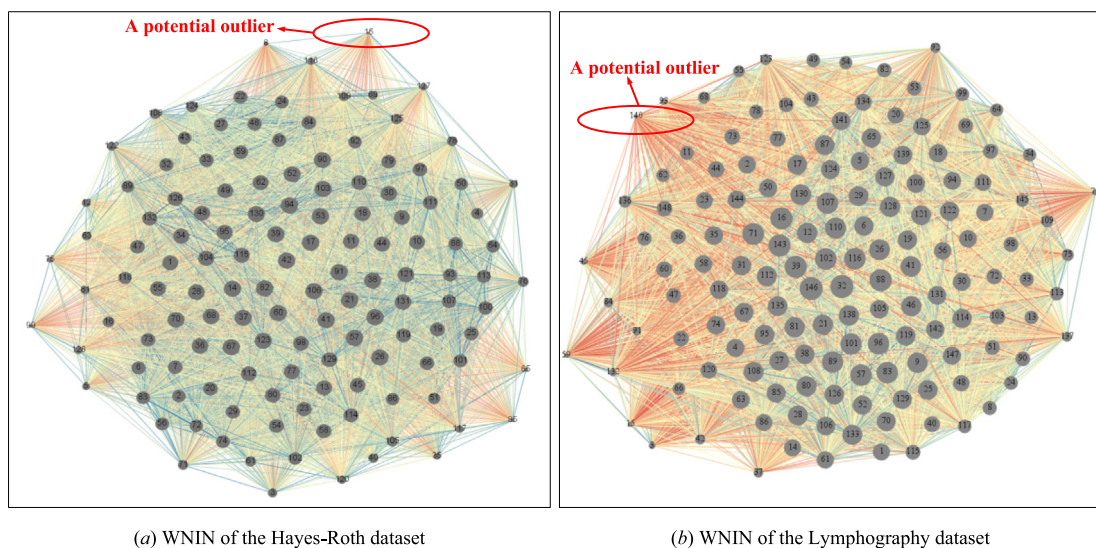


Fig. 12. The WNIN of the UCI datasets.

Table 9
Summary of mixed-valued attributes.

Attribute Number	Attribute Name	Attribute Type	Attribute Range
c ¹	Storage volume per unit time	Numeric	[0, ∞]
c ²	Employee workload	Numeric	[0, 1]
c ³	Qualification rate of quality inspections	Numeric	[0, 1]
c ⁴	Number of device types invoked	Numeric	[0, 8]
c ⁵	Accuracy of information	Numeric	[0, 1]
c ⁶	Equipment conformity	Categorical	Fully consistent A
			Fully consistent B
			High coincidence C
			Basic conformity D

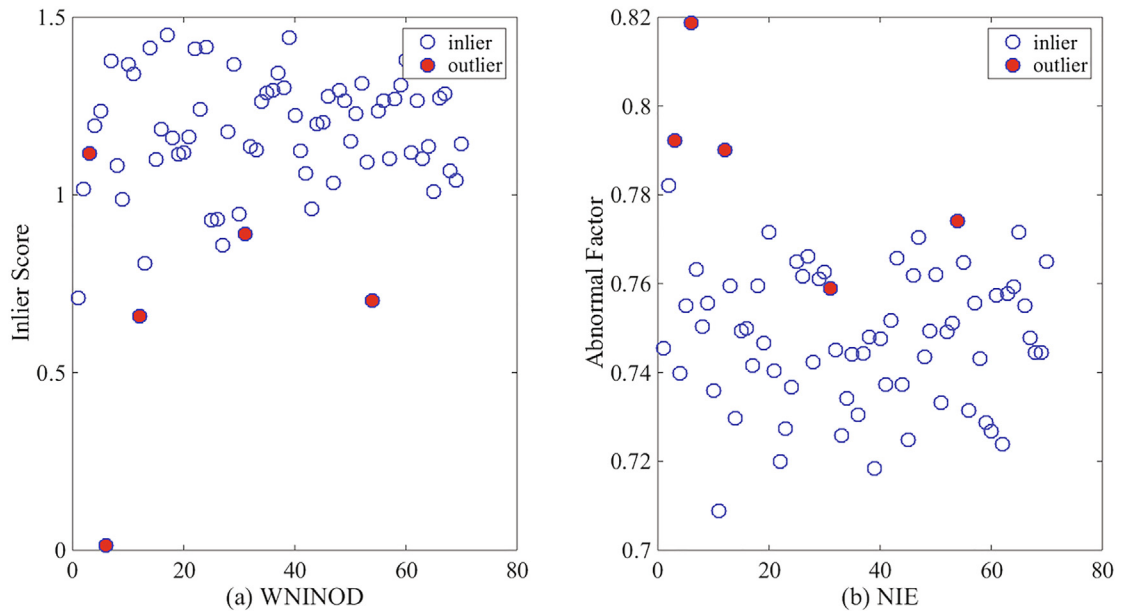


Fig. 13. Detection results of the two outlier detection methods for the real-world application.

Table 10
Comparison analysis of the two outlier detection methods for the real-world application.

Top ratio (Number of objects)	Number of rare classes included (Coverage (%))	
	WNINOD	NIE
4.3% (3)	3 (60)	3 (60)
7.1% (5)	3 (60)	4 (80)
10% (7)	4 (80)	4 (80)
25.6% (20)	5 (100)	4 (80)
34.3% (24)	5 (100)	5 (100)
Mean	4	4
Running time (s)	0.891	6.567

- (2) When the top ratio is 7.1% (the number of objects is 5), 3 (coverage 60%) and 4 (coverage 80%) actual outliers are identified by the proposed WNINOD and the NIE, respectively. When the top ratio is 25.6% (the number of objects is 20), 5 (coverage 100%) and 4 (coverage 80%) actual outliers are identified by using the two methods, respectively.
- (3) The WNINOD identified all the actual outliers when the top ratio was 7.1% (the number of objects was 20) while the NIE identified all the actual outliers when the top ratio was 34.3% (the number of objects was 24).
- (4) The running times of the proposed WNINOD and the NIE are 0.891 s and 6.567 s, respectively.

Apparently, the WNINOD has some advantages over the NIE in effectiveness and efficiency. Additionally, the iterative parameter adjustment process for the WNINOD is also relatively easier than the NIE which contains two parameters (a radius adjustment parameter λ and a judgment criterion μ) in the practical application. Therefore, the proposed method can be applied to similar real-world applications, and it even has the potential for real-time monitoring because of its high calculation efficiency.

7. Conclusions

In this paper, we proposed a new outlier detection method based on the neighbourhood rough set and Markov random walk to analyse datasets with mixed-valued attributes without any prior knowledge. The WNIN that represents the DMA is proposed using the neighbourhood relations and similarities among the objects, which guarantees strong connectivity. Outlier detection is implemented by using the tailored Markov random walk process. After the Markov random walk process reaches the equilibrium state, the stationary distribution vector is used to quantify the inlier degree of the objects. In the previous sections, an illustrative example, experimental analysis of two UCI datasets with four existing methods, and a real case study of a warehouse system's operating status were described to verify the feasibility and availability of the proposed approach. The results showed that the proposed method has better performance on the outlier detection of datasets with a single type of attribute or mixed-valued attributes. Furthermore, it may also be used to address other data mining problems such as health monitoring and binary classification.

Calculating the stationary distribution vector of the Markov random walk is a time-consuming process. Thus, future research may consider ways to overcome this specific issue and attempt to develop a new algorithm that can accelerate the solution process of the Markov stationary distribution vector while maintaining high performance. Besides, the radius adjustment parameter θ might be automatically determined by an appropriate optimization algorithm according to the real-world application.

CRedit authorship contribution statement

Yu Wang: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Yupeng Li:** Conceptualization, Methodology, Formal analysis, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

In this section, we prove the convergence of Markov random walk process based on the WNIN.

Theorem 1. *When applying Markov random walk process on a weighted neighbourhood information network, it will eventually converge to a unique stationary distribution, regardless of the value of θ .*

Proof. For a given dataset, the number of objects is fixed and finite (that is, the transition state is limited). The state transition probability matrix $\mathbf{P}$ keeping invariant across the whole process is calculated by normalizing the adjacent matrix $\mathbf{A}$, which can ensure that $p_{ij} \geq 0$ and $\forall i: \sum_j p_{ij} = 1$.

Let vector $\mathbf{V}_n = [1, \dots, 1]$, then

$$\mathbf{P} \times \mathbf{V}_n^T = \begin{bmatrix} p_{11} + p_{12} + \dots + p_{1n} \\ p_{21} + p_{22} + \dots + p_{2n} \\ \vdots \\ p_{n1} + p_{n2} + \dots + p_{nn} \end{bmatrix} = \mathbf{V}_n^T,$$

and $\mathbf{V}_n \times \mathbf{P}^T = [p_{11} + p_{12} + \dots + p_{1n}, p_{21} + p_{22} + \dots + p_{2n}, \dots, p_{n1} + p_{n2} + \dots + p_{nn}] = \mathbf{V}_n$.

$\lambda = 1$ is the eigenvalue of $\mathbf{P}^T$. Since the eigenvalues of $\mathbf{P}$ and $\mathbf{P}^T$ are the same, $\lambda = 1$ is the eigenvalue of the state transition probability matrix $\mathbf{P}$.

$$\mathbf{P}^2 = \begin{bmatrix} p_{11}p_{11} + p_{12}p_{21} + \dots + p_{1n}p_{n1} & p_{11}p_{12} + p_{12}p_{22} + \dots + p_{1n}p_{n2} & \dots & p_{11}p_{1n} + p_{12}p_{2n} + \dots + p_{1n}p_{nn} \\ p_{11}p_{11} + p_{12}p_{21} + \dots + p_{1n}p_{n1} & p_{11}p_{12} + p_{12}p_{22} + \dots + p_{1n}p_{n2} & \dots & p_{11}p_{1n} + p_{12}p_{2n} + \dots + p_{1n}p_{nn} \\ \vdots & \vdots & \ddots & \vdots \\ p_{11}p_{11} + p_{12}p_{21} + \dots + p_{1n}p_{n1} & p_{11}p_{12} + p_{12}p_{22} + \dots + p_{1n}p_{n2} & \dots & p_{11}p_{1n} + p_{12}p_{2n} + \dots + p_{1n}p_{nn} \end{bmatrix}$$

In the $\mathbf{P}^2$ process, the sum of any row vectors is $\sum_{j=1}^n p_{ij}^2 (\forall i \in (1, 2, \dots, n)) = (p_{i1}p_{11} + p_{i2}p_{21} + \dots + p_{in}p_{n1}) + (p_{i1}p_{12} + p_{i2}p_{22} + \dots + p_{in}p_{n2}) + \dots + (p_{i1}p_{1n} + p_{i2}p_{2n} + \dots + p_{in}p_{nn}) = p_{i1}(p_{11} + p_{12} + \dots + p_{1n}) + p_{i2}(p_{11} + p_{12} + \dots + p_{1n}) + \dots + p_{in}(p_{11} + p_{12} + \dots + p_{1n}) = p_{i1} + p_{i2} + \dots + p_{in} = 1$, where, $p_{ij}^2 \geq 0$, and $\forall i: \sum_j p_{ij}^2 = 1$, $\mathbf{P}^2$ is also a Markov matrix. It can be deduced that $\mathbf{P}^n$ is also a Markov matrix, and $0 < p_{ij}^n < 1$.

Suppose that $|\lambda| > 1$ is the eigenvalue of $\mathbf{P}$.

Let $\mathbf{u} = \mathbf{P}^n \times \mathbf{x} = \lambda^n \times \mathbf{x}$, then, $\mathbf{u} \rightarrow \infty$. $\mathbf{P}^n \times \mathbf{x}$ cannot be infinite because that $p_{ij}^n < 1$. This is contrary to the hypothesis, so the hypothesis is not tenable.

In conclusion, the maximum eigenvalue of the state transition probability matrix $\mathbf{P}$ is 1. With the increase of n , the items with $|\lambda| < 1$ tend to be 0, and only those with eigenvalue 1 are left. Therefore, the Markov process based on the neighbourhood information network must converge to a unique stationary distribution. $\square$

References

- [1] C.C. Aggarwal, P.S. Yu, Outlier detection for high dimensional data, *ACM Sigmod Record* 30 (2) (2001) 37–46, <https://doi.org/10.1145/376284.375668>.
- [2] S.D. Bay, The UCI KDD repository, 1999. <http://kdd.ics.uci.edu..>
- [3] M. Bouguessa, A practical outlier detection approach for mixed-attribute data, *Expert Syst. Appl.* 42 (2015) 8637–8649, <https://doi.org/10.1016/j.eswa.2015.07.018>.
- [4] H. Chen, T. Li, Y. Cai, C. Luo, H. Fujita, Parallel attribute reduction in dominance-based neighborhood rough set, *Inf. Sci.* 373 (2016) 351–368, <https://doi.org/10.1016/j.ins.2016.09.012>.
- [5] Y. Chen, D. Miao, H. Zhang, Neighborhood outlier detection, *Expert Syst. Appl.* 37 (12) (2010) 8745–8749, <https://doi.org/10.1016/j.eswa.2010.06.040>.
- [6] Y. Chen, Z. Zhang, J. Zheng, Y. Ma, Y. Xue, Gene selection for tumor classification using neighborhood rough sets and entropy measures, *J. Biomed. Inform.* 67 (2017) 59–68, <https://doi.org/10.1016/j.jbi.2017.02.007>.
- [7] T.M. Cover, P.E. Hart, Nearest neighbor pattern classification, *IEEE Trans. Inf. Theory* 13 (1) (1967) 21–27, <https://doi.org/10.1109/tit.1967.1053964>.
- [8] T.M. Cover, J.A. Thomas, Elements of information theory, *Publ. Am. Statist. Assoc.* 103 (481) (1992) 429, <https://doi.org/10.1198/jasa.2008.s218>.
- [9] M. Fort, J.A. Sellares, Finding influential location regions based on reverse k-neighbor queries, *Knowl.-Based Syst.* 47 (jul.) (2013) 35–52, <https://doi.org/10.1016/j.knsys.2013.03.013>.
- [10] Geyer, J. Charles, Practical markov chain monte carlo, *Stat. Sci.* 7 (4) (1992) 473–483, <https://doi.org/10.1214/ss/1177011137>.
- [11] P.Y. Goh, S.C. Tan, W.P. Cheah, Anomaly detection using correctness matching through a neighborhood rough set, in: *International Conference on Neural Information Processing*, Springer, Cham, 2016, pp. 434–441, https://doi.org/10.1007/978-3-319-46675-0_47.
- [12] P.Y. Goh, S.C. Tan, W.P. Cheah, Mining outliers from medical datasets using neighborhood rough set and data classification with neural network, in: *Emerging Trends in Neuro Engineering and Neural Computation*, Springer, Singapore, 2017, pp. 219–228, https://doi.org/10.1007/978-981-10-3957-7_12.
- [13] D.M. Hawkins, *Identification of outliers*, Chapman and Hall, London, 1980.
- [14] R.A. Horn, The hadamard product, *Proc. Symp. Appl. Math* 40 (1990) 87–169, <https://doi.org/10.1090/psapm/040/1059485>.
- [15] J. Ji, T. Bai, C. Zhou, C. Ma, Z. Wang, An improved k-prototypes clustering algorithm for mixed numeric and categorical data, *Neurocomputing* 120 (10) (2013) 590–596, <https://doi.org/10.1016/j.knsys.2012.01.006>.
- [16] E.M. Knorr, R.T. Ng, Algorithms for mining distance-based outliers in large datasets, in: *Proceedings of the 24th VLDB Conference*, 1998, pp. 392–403.
- [17] N. Li, C. Zhang, W. Xie, Y. Li, Exceptional events classification in warehousing based on an integrated clustering method for a dataset with mixed-valued attributes, *Int. J. Comput. Integr. Manuf.* 31 (11) (2018) 1–19, <https://doi.org/10.1080/0951192X.2018.1509129>.
- [18] W. Li, W. Mo, X. Zhang, J.S. John, Y. Lu, W.S. Eric, W. Fan, J.M. DiMaio, E.T. Jeffrey, Outlier detection and removal improves accuracy of machine learning approach to multispectral burn diagnostic imaging, *J. Biomed. Opt.* 20 (12) (2015), <https://doi.org/10.1117/1.JBO.20.12.121305>.
- [19] X. Li, S. Tian, T. Qiu, Y. Chen, Outlier detection using the information entropy of neighborhood rough sets, *J. Inf. Comput. Sci.* 9 (12) (2012) 3339–3350.
- [20] Y. Li, Y. Wang, N. Li, Abnormal operation status identification in warehousing based on neighborhood information entropy considering mixed-valued attributes, *Int. J. Prod. Res.* (2020), <https://doi.org/10.1080/00207543.2020.1788736>.
- [21] Y. Li, D. Wu, J. Ren, C. Hu, An improved outlier detection method in high-dimension based on weighted hypergraph, in: *2009 Second International Symposium on Electronic Commerce and Security*, IEEE, 2009, pp. 159–163.
- [22] Z. Li, L. Chen, Z. Ren, S.S. Tirumala, A pattern recognition for group abnormal behaviors based on markov random fields energy, in: *2014 IEEE 13th International Conference on Cognitive Informatics and Cognitive Computing*, IEEE, 2014, pp. 526–528.
- [23] Y. Liu, L. Zhang, L. Nie, Y. Yan, D.S. Rosenblum, Fortune teller: predicting your career path, *National conference on artificial intelligence* (2016) 201–207, <https://doi.org/10.5555/3015812.3015842>.
- [24] H.D.K. Moonesignhe, P.N. Tan, Outlier detection using random walks, in: *2006 18th IEEE International Conference on Tools with Artificial Intelligence*, IEEE, 2006, pp. 532–539.
- [25] John Odentantz, Markov chains: Gibbs fields, monte carlo simulation, and queues, *Technometrics* 42 (4) (2010) 438–439, <https://doi.org/10.1080/00401706.2000.10485734>.
- [26] M.E. Otey, A. Ghoting, S. Parthasarathy, Fast distributed outlier detection in mixed-attribute data sets, *Data Min. Knowl. Discovery* 12 (2–3) (2006) 203–228, <https://doi.org/10.1007/s10618-005-0014-6>.
- [27] R.S. Prasad, S.M. Nejres, Use of data mining techniques for weather data in basra city, *Int. J. Adv. Res. Comput. Sci. Software Eng.* 5 (12) (2015) 135–139.
- [28] D. Preotiucpietro, Y. Liu, D.J. Hopkins, L.H. Ungar, Beyond binary labels: political ideology prediction of twitter users, in: *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, 2017, pp. 729–740.
- [29] P.J. Rana, J. Baria, A survey on fraud detection techniques in ecommerce, *Int. J. Comput. Appl.* 113 (14) (2015) 5–7, <https://doi.org/10.5120/19892-1898>.
- [30] S. Ranshous, S. Shen, D. Koutra, S. Harenberg, C. Faloutsos, N.F. Samatova, Anomaly detection in dynamic networks: a survey, *Wiley Interdiscip. Rev. Comput. Stat.* 7 (3) (2015) 223–247, <https://doi.org/10.1002/wics.1347>.

- [31] T. Reuter, P. Cimiano, Event-based classification of social media streams, International conference on multimedia retrieval (2012) 1–18, <https://doi.org/10.1145/2324796.2324824>.
- [32] L. Sun, X. Zhang, J. Xu, W. Wang, R. Liu, A gene selection approach based on the fisher linear discriminant and the neighborhood rough set, Bioengineered 9 (2017) 144–151, <https://doi.org/10.1080/21655979.2017.1403678>.
- [33] C. Wang, H. Lin, H. Jiang, Trajectory-based multi-dimensional outlier detection in wireless sensor networks using hidden markov models, Wireless Netw. 20 (8) (2014) 2409–2418, <https://doi.org/10.1007/s11276-014-0757-3>.
- [34] C. Wang, H. Gao, Z. Liu, Y. Fu, A new outlier detection model using random walk on local information graph, IEEE Access 6 (2018) 75531–75544, <https://doi.org/10.1109/ACCESS.2018.2883681>.
- [35] C. Wang, H. Gao, Z. Liu, Y. Fu, Outlier detection using diverse neighborhood graphs, in: 2018 15th International Computer Conference on Wavelet Active Media Technology and Information Processing (ICCWAMTIP), IEEE, 2018, pp. 58–62.
- [36] C. Wang, Z. Liu, H. Gao, Y. Fu, Vos: A new outlier detection model using virtual graph, Knowl.-Based Syst. 185 (2019) 104907.1–104907.12, <https://doi.org/10.1016/j.knosys.2019.104907>.
- [37] X. Wang, I. Davidson, Discovering contexts and contextual outliers using random walks in graphs, in: 2009 Ninth IEEE International Conference on Data Mining, IEEE, 2009, pp. 1034–1039.
- [38] W. Wei, J. Liang, Y. Qian, A comparative study of rough sets for hybrid data, Inf. Sci. 190 (2012) 1–16, <https://doi.org/10.1016/j.ins.2011.12.006>.
- [39] D.R. Wilson, T.R. Martinez, Improved heterogeneous distance functions, J. Artif. Intell. Res. 11 (1) (1997) 1–34, <https://doi.org/10.1109/ISIE.1997.648935>.
- [40] H.N. Win, K.T. Lynn, Community and outliers detection in social network, in: International Conference on Big Data Analysis and Deep Learning Applications, Springer, Singapore, 2018, pp. 58–67.
- [41] C. Zhang, A. Yin, Anomaly detection algorithm based on subspace local density estimation, Int. J. Web Serv. Res. 16 (3) (2019) 44–58, <https://doi.org/10.4018/IJWSR.2019070103>.
- [42] Y. Zhong, Z. Xianrong, F. Shan, Hybrid data-driven outlier detection based on neighborhood information entropy and its developmental measures, Expert Syst. Appl. 112 (2018) 243–257, <https://doi.org/10.1016/j.eswa.2018.06.013>.
- [43] P. Zhou, X. Hu, P. Li, X. Wu, Online streaming feature selection using adapted neighborhood rough set, Inf. Sci. 481 (2019) 258–279, <https://doi.org/10.1016/j.ins.2018.12.074>.
- [44] Q. Zhu, J. Feng, J. Huang, Weighted natural neighborhood graph: an adaptive structure for clustering and outlier detection with no neighborhood parameter, Cluster Comput. 19 (3) (2016) 1–13, <https://doi.org/10.1007/s10586-016-0598-1>.